



Split+

Split parts, surfaces, curves or faces with several tools. Optionally supply reference geometry from which we can establish a more stable definition of which side of the geometry to split we intend to keep or remove

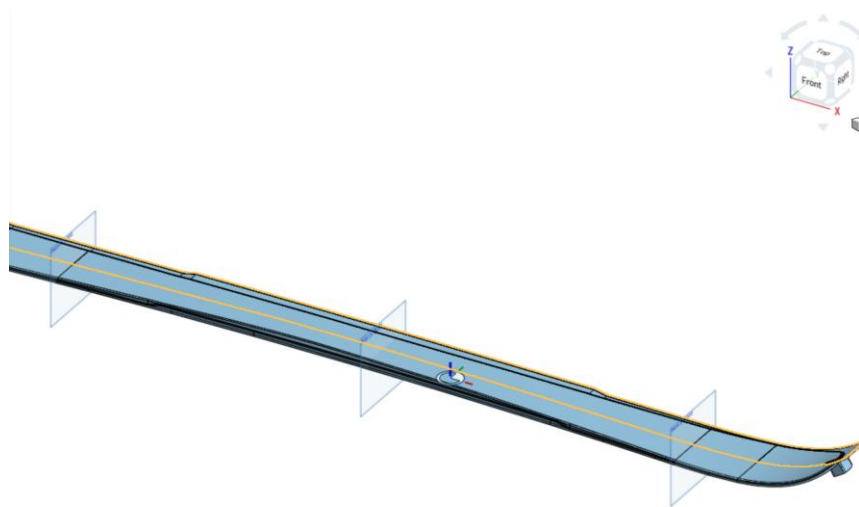


What it does

The built-in Split keeps "front" or "back" of one tool. That choice is tied to the tool's normal, which flips whenever an upstream loft, offset, mirror or sketch changes. Split+ splits by many tools at once and names the side to keep with geometry: a reference that lies on that side.

Use it when

- A body must be cut by several planes or surfaces in one step
- You keep one side, and the tools' normals may change upstream
- Later features need the pieces by position: start, middle, end



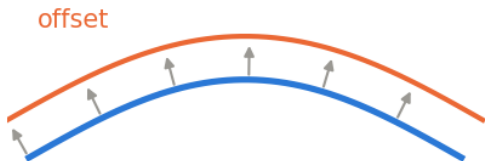
Split+ open on a ski topsheet: the target and the two tools (Station A and B) highlight.



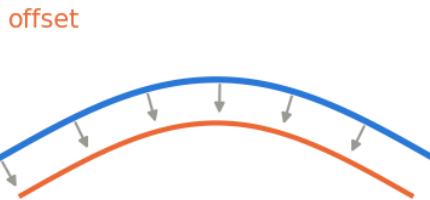
Why a reference instead of a flip

A flip names a side relative to a normal; a reference names it in the model

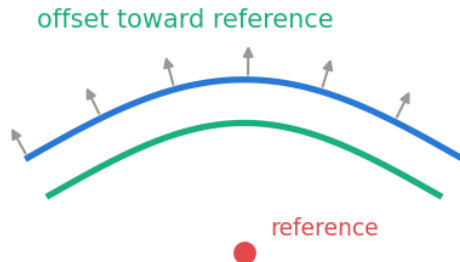
Built-in, normal up:
'offset along normal' goes up



Same model after an upstream change
flips the normal: now it goes DOWN



Offset+ 'toward reference':
goes toward the point, either way



A flip is about a normal

'Keep front' means the side the tool's normal points to -- and normals change freely upstream.

A reference is about the model

Pick geometry on the side you mean. The choice survives any change of orientation.

Drop-in

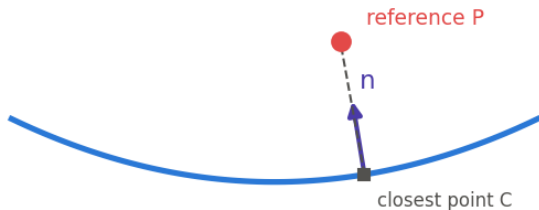
With no reference, Split+ behaves exactly like the built-in Split.



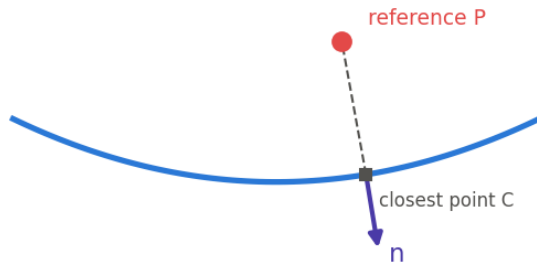
How the side is decided

The side is read, not assumed: the sign of s says where the reference is for THIS normal

Normal up
 $s = (P - C) \cdot n = +31 \rightarrow$ reference on the FRONT



Normal down
 $s = (P - C) \cdot n = -31 \rightarrow$ reference on the BACK



Signed distance

For each tool: $s = (P - C) \cdot n$ at the closest point. $s > 0$: the reference is in front of the tool.

Per tool

Each tool keeps the reference's side (box on) or the other side (box off). What survives is on the reference's side of every tool.

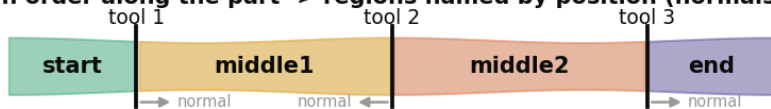
No reference

Behaves like the built-in: on = each tool's front, off = its back.

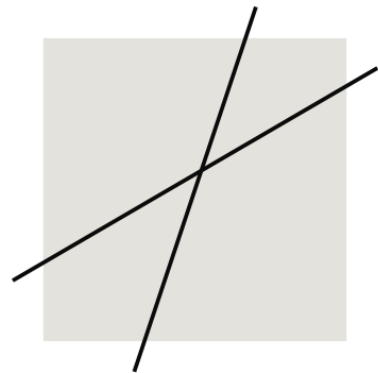


Regions are named by position

Tools picked in order along the part -> regions named by position (normals don't matter)



No bands -> no regions



crossing tools: the split is made,
region keys published EMPTY + info notice

Pick tools in order

1 tool: near / far (or front / back). 2 tools: start, middle, end. 3+: start, middle1..N-1, end.

Orientation never matters

'Forward' for each tool is the side the next tool is on.

Crossing tools

The split is made as asked; region keys are published empty, with an info notice.



The dialog: split, keep both sides

Split+ topsheet into 3 reg... ✓ ✗

1 **Part** Face

2 Parts, surfaces, or curves to split
Filletted_Top_Surf ✕

3 Entities to split with ⌚
Station A (x 300) ✕
Station B (x 1500) ✕

4 ☐ Keep tools

5 ☐ Trim to face boundaries

6 ☒ Keep both sides

7 > Debug

Final

#	Parameter	What it means
1	Split type	PART: split whole bodies. FACE: split only the picked faces; nothing is removed.
2	Parts, surfaces, or curves to split	The targets. Every target is split by every tool.
3	Entities to split with	Surfaces, faces, planes or mate connectors, in pick order. Pick them in order along the part: region names follow it.
4	Keep tools	Keep sheet-body tools after the split.
5	Trim to face boundaries	A single-face tool cuts only within its own boundary, as the built-in.
6	Keep both sides	On (default): every piece is kept and sorted into regions. Off: keep one side per tool (next slide).
7	Debug	Print sides: which side of each tool the reference is on, and what was kept.



The dialog: keep one side

Split+ volume, keep betw... ✓ ✗

Part Face

Parts, surfaces, or curves to split
VOLUME_STD ✕

Entities to split with 🕒
Station A (x 300) ✕
Station B (x 1500) ✕

☐ Keep tools
☐ Trim to face boundaries
☐ Keep both sides

Side reference 🕒 📏
Mid (x 900) ✕

> Debug

Final

#	Parameter	What it means
1	Keep both sides	Off: each tool keeps one side.
2	Side reference	Geometry on the side to keep -- best a mate connector or vertex inside the region you want. Here: Mid (x 900).
3	Keep side nearest reference	The arrow. On: keep the side of each tool the reference is on. Off: keep the other side. Without a reference: each tool's front / back.

Result: only the region between Station A and B survives, whatever way the two planes face (example 2).



The dialog: face split

Split+ faces: SW_SHELF ... ✓ ✗

1 Part Face

2 Faces to split (Q)
Face of Derived 1 ✕

Entities to split with (Q)
Station A (x 300) ✕
Station B (x 1500) ✕

☒ Keep tools

3 Side reference (Q)

> Debug

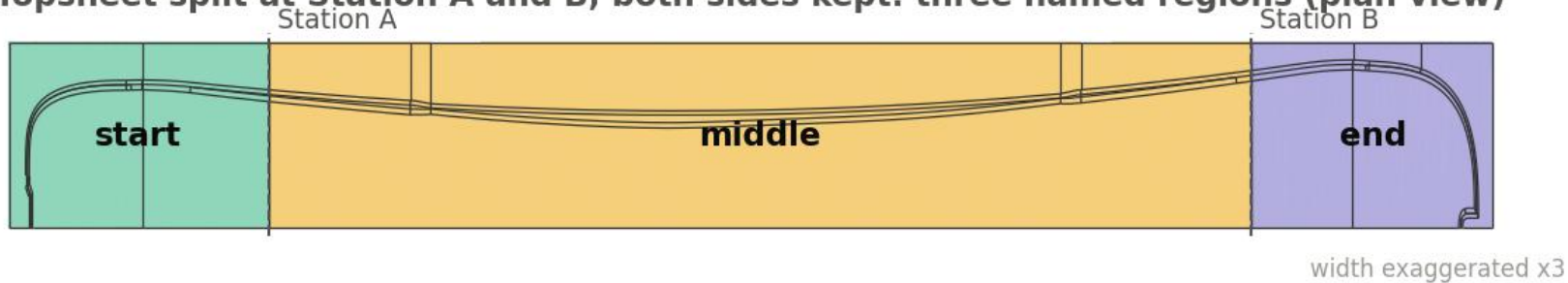
#	Parameter	What it means
1	Split type	FACE: split only the picked faces. Nothing is removed; the pieces stay part of their body.
2	Faces to split	The faces to split. Planes and mate connectors cut them infinitely; surfaces cut as they are.
3	Side reference	Optional. With one tool it names the two regions near / far; it never removes anything here.

Regions are published as faces: pick 'middle' in Extract variables to fillet or offset only those faces.



Example 1: three regions

Topsheet split at Station A and B, both sides kept: three named regions (plan view)



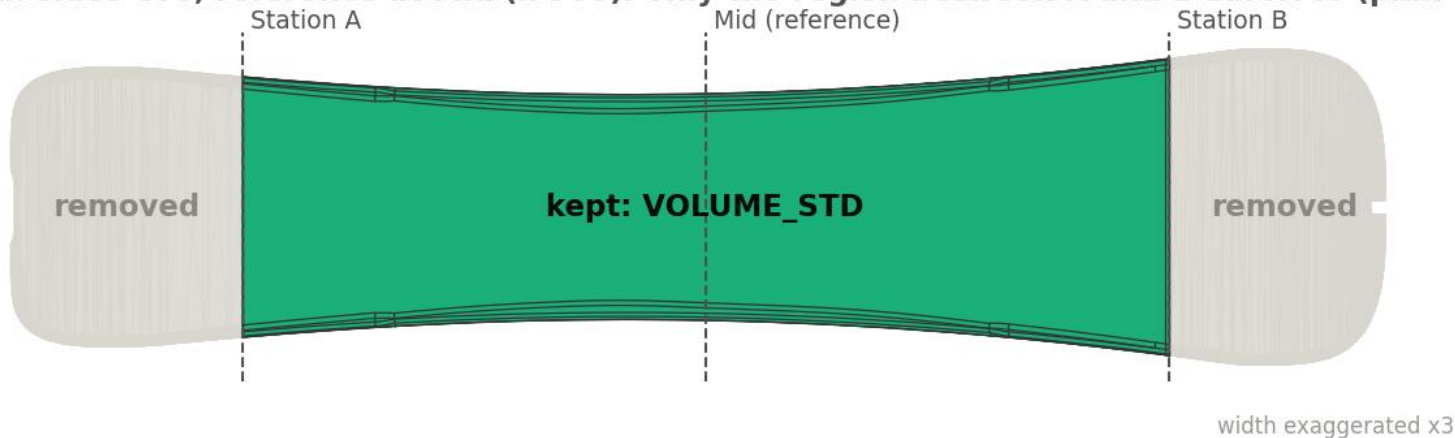
Topsheet split at Station A (x 300) and B (x 1500), Keep both sides on

Nothing is removed. The three pieces are published as start (x -19..300), middle (300..1500) and end (1500..1799), and the cut edges as splitEdges -- one edge per cut: 9 edges, 5 at A and 4 at B.



Example 2: keep only the middle

Keep both sides OFF, reference at Mid (x 900): only the region between A and B survives (plan view)



VOLUME_STD split at Station A and B, Keep both sides off, Side reference = Mid (x 900)

Each station plane keeps the side Mid is on, so only the part between A and B survives (green); the tip and tail pieces (grey) are removed. Flipping either plane's normal changes nothing -- which the built-in 'keep front / back' cannot promise.



Example 3: split faces, not bodies

Face split of the SW_SHELF faces at A and B: nothing removed, faces sorted into regions (side view)



Face split: the SW_SHELF faces (the shelf along the sidewall) cut at Station A and B

Nothing is removed and no new body is made: the shelf faces are divided at A and B and published as regions -- start 4 faces, middle 9, end 4. A later feature (a fillet, an offset, a Thicken+) can then work on 'middle' alone, and keeps working when the stations move.



Outputs: what Extract variables can publish

Published for Extract variables

Key	What it holds
output	Every resulting piece.
near / far, start / middle* / end	The pieces in each region: bodies (solids, surfaces or curves) in a Part split, faces in a Face split.
<region>Edges	A region's edges except the cuts (for a surface: its open boundary).
outside / inside (+ Edges)	start + end together / every middle region together.
splitEdges	The NEW edges along each cut line -- where a tool crosses a target. Keeping both sides leaves two coincident copies of every cut edge; only one is published. Input edges that were merely cut in two are not included.
cut, startCut / endCut, cut1..N	splitEdges sorted by the tool that made them (each edge goes to its nearest tool): 1 tool: cut; 2 tools: startCut, endCut; 3+: cut1..cutN. Topsheet example: startCut = the 5 edges at x 300.
splitFaces	The cap faces the split created on solids (empty for surfaces and in a Face split).
pieceCount, regionCount	Numbers: pieces left; regions the tools define (N + 1 for N tools).



Messages

Messages

	Message / what to do
Error	A tool is also a target Pick it only once.
Error	The keep side reference lies on tool k Move the reference clearly to one side of that tool.
Error	Nothing is left No part of the targets is on the kept side of every tool.
Info	Split done; regions not published: ... Tools cross, or were picked out of order. The split itself is fine.



Tips and limits



Use a local reference

A mate connector or vertex inside the region to keep. A far plane can read the wrong side where the part curves.



Pick tools along the part

Region names follow pick order.



Reference regions, not indices

Extract variables can publish 'middle' or 'startCut'; those names survive changes upstream.



Outputs keep their input's orientation

The reference picks the side to keep, not the normal: pieces face the way the input faced (the topsheet example faces down). Built-ins downstream (Thicken, Offset surface, Move face) follow that normal -- put Orient to reference before them, or use Thicken+ / Offset+.



Edge tools are not offered

The side rule needs a surface. Use the built-in Split for projected edges.



How pieces are sorted into regions

Split+ checks which side of each tool ONE point of each piece lies on (its centre; for a face, the nearest point on the face). A piece that curls back across a tool's extended plane could land in the wrong region -- check regions with Show in Extract variables.



Where it lives

Document	Reference_Side_Features (22764764a00a7f607dbc1c4d)
Tab	split_plus (uses reference_side_utils)
Tests	Reference side tests studio: S1-S10 - devtools/onshape/check_reference_side_tests.py
Examples	topsheet_surf studio: Split+ topsheet into 3 regions, Split+ volume keep between A and B, Split+ faces
Explainer	docs/explainers/reference_side/reference_side_explained.md, section 2.2
Related	Mutual Trim+, Offset+, Thicken+ (same reference rule); Extract variables